

COMEX SILVER FUTURES (SI)

Spread & Curve Structure — Fundamental Reference

The Dual-Identity Metal: Monetary Asset and Industrial Commodity

This document presents a fact-based reference on the fundamental drivers of COMEX Silver futures (ticker: SI), contract size 5,000 troy ounces, with particular focus on the forces that shape the forward curve and the spread structure between contract months. Silver occupies a distinctive and somewhat unstable position among major commodities: it shares gold's monetary, store-of-value, and safe-haven characteristics — silver has a multi-millennia history as a monetary metal and remains a central bank-adjacent and retail-investment asset today — while simultaneously functioning as a genuine industrial commodity with substantial and growing consumption in electronics, solar photovoltaics, and other manufacturing applications that have no parallel in the gold market. This dual identity means silver responds to the real-interest-rate and USD framework that dominates gold pricing while also responding to industrial demand cycles, manufacturing PMI data, and a structural deficit narrative tied to the energy transition that has more in common with copper than with gold. The result is a metal with higher volatility than gold (silver's price moves are conventionally understood in the trade to run at a multiple of gold's on a percentage basis) and a forward curve whose carry structure can, at times, deviate from a pure interest-rate framework when industrial-demand-driven physical tightness asserts itself. Understanding both halves of this dual identity — and the gold-silver ratio that is used to track the balance between them — is the foundation of spread and butterfly trading on the SI curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and the Physical Silver Market

COMEX Silver futures are the primary US exchange for silver price discovery and risk transfer, operating alongside the London OTC silver market in a structure broadly analogous to — though with some distinct features from — the gold market described in the separate COMEX Gold reference document in this series.

Parameter	Detail
Exchange	CME Group — COMEX division (New York)
Ticker	SI (front month); a smaller-sized "mini" contract (SIL, 1,000 oz) also trades on COMEX for participants seeking reduced contract size
Contract size	5,000 troy ounces
Quotation	US cents per troy ounce; minimum tick = \$0.005/oz = \$25.00 per contract
Contract months	January, March, May, July, September, December as the primary actively quoted months (with additional nearby months listed); active contracts extend approximately 2-3 years forward. Liquid concentration in front 3-5 contracts.
Settlement type	Physical delivery of silver bars (1,000 troy oz standard bars, minimum 99.9% fineness, COMEX-approved brands/refiners) at COMEX-approved depositories in the New York/New Jersey area. As with gold, the substantial majority of contracts are cash-settled or rolled rather than taken to physical delivery.
Trading hours	CME Globex: Sunday-Friday 18:00-17:00 ET (23-hour session); liquid hours approximately 08:00-13:25 ET (overlapping the LBMA London session)
Last trading day	Third-to-last business day of the delivery month
Contract value at \$30/oz	\$150,000 per contract
LBMA relationship	The London Bullion Market Association (LBMA) runs the global OTC silver market in parallel with its gold market, and publishes a daily LBMA Silver Price (a single daily auction-based fix, administered by ICE Benchmark Administration, in contrast to gold's twice-daily AM/PM fix structure). The COMEX-LBMA spread (the silver EFP) is analyzed analogously to the gold EFP described in the COMEX Gold reference document.

1.1 Why Silver's Contract Structure Differs Slightly From Gold's

- Silver's materially lower per-ounce price than gold means a COMEX silver contract (5,000 oz) is sized differently in physical-ounce terms than a gold contract (100 oz) in order to produce a broadly comparable contract notional value — though silver's notional value per contract still moves with its own price level and is not pegged to track gold's contract value over time.
- **The "mini" silver contract (SIL, 1,000 oz)** exists alongside the standard 5,000 oz contract to provide a smaller-notional instrument; gold has an analogous smaller contract (the COMEX miNY Gold contract) for the same reason — accommodating participants for whom the standard contract size represents more notional exposure than desired.

1.2 The LBMA Silver Price — A Single Daily Fix

- Unlike gold's twice-daily (AM and PM) LBMA fixing structure, the LBMA Silver Price is set once per day via an electronic auction process administered by ICE Benchmark Administration (the same

administrator used for the gold fix). This is a structural, documented difference in the benchmark-setting process between the two metals' London reference prices, though both serve the same general function as the primary physical price reference used for contracts, ETF NAV calculations, and other settlement purposes.

2. The Dual Identity — Why Silver Behaves Differently From Gold

The single most important conceptual starting point for analysing silver is that, unlike gold, it cannot be modelled primarily as a monetary asset. Silver's annual mine and recycled supply relative to its above-ground stock, and the share of that annual supply absorbed by industrial fabrication rather than investment or jewellery, place it in a meaningfully different analytical category from gold.

2.1 The Stock-to-Flow Distinction From Gold

- As described in the COMEX Gold reference document, gold's total above-ground stock is approximately 60–70 times annual mine production, meaning annual production flow is a small fraction of the total stock and has limited short-term price impact. Silver's stock-to-flow ratio is materially lower than gold's: a substantial share of historical above-ground silver stock has been consumed by industrial and photographic uses over the decades (uses that, unlike with gold, are generally not recovered and re-circulated to the same degree, since silver is often used in small, dispersed quantities in electronic components and other applications where full recovery is not economic). This means silver's available above-ground investment-grade stock relative to annual supply and demand flow is structurally tighter than gold's, a factor cited by analysts as contributing to silver's higher price volatility.
- **Practical implication:** because a meaningfully larger share of annual silver supply is consumed (industrially) rather than held (as an investment or monetary asset), shifts in industrial demand have a proportionally larger effect on the silver supply-demand balance than equivalent shifts in jewellery or industrial gold demand have on the gold balance — and this is the core reason silver requires an industrial-commodity analytical framework in addition to gold's monetary-asset framework.

2.2 When Each Framework Dominates

- **The monetary/real-rate framework tends to dominate** during periods of major monetary policy shifts, financial-system stress, and broad safe-haven flows — silver generally moves in the same direction as gold during these episodes, often with greater percentage amplitude (see Section 9 on the gold-silver ratio).
- **The industrial-demand framework tends to dominate** during periods of significant divergence in global manufacturing activity, solar installation growth, or other industrial-demand shocks that do not have an equivalent effect on gold — these are the periods most likely to produce silver-specific price moves that diverge from gold, and are the periods in which the gold-silver ratio moves most significantly.
- In most periods, both frameworks operate simultaneously to some degree, and a silver price move's underlying driver is best assessed by checking it against both the real-rate/USD framework (Section 3) and the industrial demand and inventory data (Sections 4–6) rather than assuming either framework in isolation.

3. The Monetary and Financial Framework

To the extent silver is priced as a monetary/investment asset, it is subject to substantially the same real interest rate and USD framework described in full in the COMEX Gold reference document. This section summarises that framework as it applies to silver specifically, without duplicating the full derivation already presented for gold.

3.1 Real Interest Rates

- Silver, like gold, is a non-yielding asset, and holding it carries an opportunity cost approximated by the real interest rate (commonly referenced via the 10-year TIPS yield, as described for gold). The silver price has historically exhibited a documented positive correlation with the gold price and a corresponding negative correlation with real rates, though the relationship is generally regarded by analysts as somewhat less tight and less reliable for silver than for gold, reflecting the industrial-demand component that gold does not share.

3.2 USD and Safe-Haven Flows

- The US Dollar Index (DXY) negative correlation mechanism described for gold in Section 9 of the COMEX Gold reference document applies analogously to silver, since silver is also a globally traded, USD-denominated commodity. Silver also participates in broad safe-haven flows during financial-system stress episodes, generally moving in the same direction as gold, subject to the amplification/divergence dynamics described via the gold-silver ratio in Section 9.

3.3 Retail and ETF Investment Demand

- **iShares Silver Trust (SLV):** the world's largest silver-backed ETF, the silver-market analogue to gold's GLD, publishing daily holdings that serve as a real-time investment-demand signal.
- **Retail bar and coin demand:** as with gold, silver bar and coin investment demand (particularly significant in the US, Germany, and other markets with an established retail precious metals investment culture) tends to be counter-cyclical to some degree, rising during periods of financial uncertainty.
- **The silver market's smaller absolute size relative to gold** means that a given dollar inflow or outflow from ETFs or retail investment demand represents a proportionally larger share of the total silver market than an equivalent dollar flow represents for the much larger gold market — a structural factor frequently cited by analysts as contributing to silver's higher volatility, independent of the industrial-demand factors discussed elsewhere in this document.

4. Industrial Demand — The Defining Difference From Gold

Industrial fabrication demand is the component of the silver market with no meaningful equivalent in gold, and is the primary reason silver requires its own dedicated demand-side analytical framework distinct from gold's.

Industrial Application	Approx. Share of Total Silver Demand	Key Driver	SI Curve Sensitivity
Solar photovoltaics	~15-20% and rising	Global solar panel installation growth; silver paste/conductor loading per panel (subject to ongoing thrifting efforts by panel manufacturers to reduce silver content per watt)	The most closely watched and fastest-growing industrial demand category; a structural driver underpinning recent multi-year deficit narratives (Section 8).
Electronics and electrical contacts	~25-30%	Consumer electronics production; silver's use in switches, contacts, and conductive pastes owing to its highest electrical conductivity of any metal	Tied to global electronics manufacturing and consumer demand cycles; semiconductor industry capex relevant.
Brazing alloys and solders	~5-8%	Industrial manufacturing and construction activity; HVAC and plumbing applications	Tied to broader industrial production and construction cycles.
Photography	<2%, structurally declining	Largely displaced by digital photography; a historically significant but now minor demand category	Negligible current relevance; included for completeness given its historical importance to the silver market.
Jewellery and silverware	~15-20%	India is the largest single silverware/jewellery market; price-sensitive, similarly to gold jewellery demand but at silver's lower price point	Indian silver jewellery/silverware demand is seasonally tied to the same wedding-season and festival calendar described for Indian gold demand in the COMEX Gold reference document.
Other industrial (catalysts, medical, water purification, mirrors)	~5-8% combined	Diverse smaller applications, including silver's antimicrobial properties in medical and water-treatment uses	Smaller individually but collectively meaningful; generally stable, slow-growing demand.

4.1 Solar Photovoltaics — The Structural Growth Story

- **Why solar panels use silver:** crystalline silicon solar cells (the dominant commercial solar technology) use a silver-based conductive paste to collect and transport the electrical current generated by the cell, owing to silver's combination of high electrical conductivity and suitability for the screen-printing manufacturing process used in cell production.
- **Thrifting:** solar panel manufacturers have an ongoing commercial incentive to reduce ("thrift") the silver content per panel/per watt of capacity, both to reduce input costs and to reduce exposure to silver price volatility; reported industry progress on thrifting has been significant over the past decade, partially offsetting the demand growth that would otherwise result purely from rising panel installation volumes. The net demand trajectory therefore depends on the balance between installation volume growth and the pace of continued thrifting, both of which are tracked separately

by industry analysts.

- **Alternative cell technologies:** some emerging photovoltaic cell architectures (including certain heterojunction and other next-generation designs under industry development) carry different — in some cases higher — silver loading per cell than mainstream technology, while other research and development efforts specifically target silver reduction or substitution; the net long-run effect of evolving cell technology on aggregate silver demand is, as of the time of writing, a subject of ongoing industry analysis rather than a settled trajectory.

5. Mine Supply — The By-Product Production Structure

Silver mine supply has a structural feature with no real equivalent in gold: the substantial majority of silver mine production is recovered as a by-product or co-product of mining for other base metals — primarily lead, zinc, and copper — rather than from mines whose primary economic purpose is silver production. This has important implications for how silver supply responds (or fails to respond) to the silver price specifically.

5.1 By-Product and Co-Product Supply Structure

- A substantial majority of global silver mine production (commonly cited by industry sources as roughly 70%, though the precise figure varies by year and methodology) comes from mines where silver is a by-product or co-product of base metal mining (lead-zinc mining in particular, with copper mining contributing a further share) or where silver is recovered alongside gold at primarily gold-focused mines. Only a minority of global silver supply comes from mines where silver is the primary economic product (so-called "primary silver mines").
- **The critical implication for supply elasticity:** because most silver supply is tied to the production economics of lead, zinc, copper, or gold mining, a rise in the silver price alone — without an accompanying rise in the price of the associated base metal — has limited ability to incentivise increased silver output, since the mine's primary investment and operating decisions are driven by the economics of its primary metal. This is a structurally different supply-response mechanism from gold, where (per the COMEX Gold reference document) the price of gold itself is the primary driver of mine investment decisions for the large majority of gold-focused operations.

5.2 Major Silver Producing Countries

Country	Annual Production (approx.)	Primary Mine Type	SI Relevance
Mexico	~180–200 Moz	Predominantly by-product of lead-zinc and some primary silver mines	The world's largest silver-producing country; Mexican mining sector conditions (including occasional labour and regulatory developments) are a closely tracked supply variable.
China	~110–120 Moz	Predominantly by-product of lead-zinc and other base metal mining	Significant producer; also a major industrial consumer of silver, making China relevant on both sides of the balance.
Peru	~110–120 Moz	By-product of polymetallic (lead-zinc-silver-copper) mining	Major producer; subject to the same community/social conflict and logistics risk factors described for Peruvian copper mining elsewhere in this document series.

Chile	~40–45 Moz	Predominantly by-product of copper mining	Tied closely to Chilean copper mine output and investment decisions, given the by-product relationship.
Russia, Poland, Australia, Bolivia, others	Variable, collectively significant	Mixed by-product and primary	Secondary producers; Russian supply subject to the same sanctions-related market segmentation risk described for Russian gold production in the COMEX Gold reference document.

5.3 Secondary Supply — Recycling

- Silver recycling — primarily from photographic film/processing waste (a structurally declining source), spent industrial catalysts, electronics scrap, and silverware/jewellery scrap — is a meaningful secondary supply source, though, as noted in Section 2.1, a substantial share of historically consumed industrial silver (particularly small quantities dispersed in electronic components) is not economically recoverable, distinguishing silver's overall recycling dynamics from gold's considerably higher effective recycling rate.

6. Exchange Inventories and Physical Market Signals

As with gold, COMEX warehouse stock data is a closely watched near-term physical market signal for silver, with some distinct features reflecting silver's different market structure.

6.1 COMEX Registered and Eligible Silver Stocks

- COMEX publishes daily registered (available for delivery) and eligible (meets delivery standards but not registered) silver stock data, structurally identical in reporting format to the gold stock data described in the COMEX Gold reference document.
- **The 2021 "silver squeeze" retail-driven episode:** in early 2021, a period of coordinated retail investor interest in silver (occurring in the broader context of that period's retail-investor-driven market activity in several assets) was associated with a sharp, short-term increase in demand for physical silver products (notably retail coin and small bar premiums, which rose significantly above the underlying futures/spot price for a period) and a documented surge in SLV ETF inflows, though the episode's effect on COMEX futures pricing itself was more modest and short-lived than the effect on physical retail product premiums and availability.

6.2 London Vault Holdings

- The LBMA publishes data on total silver holdings in London vaults (analogous to the gold vault holdings data described for gold), providing a broader view of global above-ground investment-grade silver inventory beyond the COMEX-specific registered/eligible figures.

7. The Forward Curve — Carry Structure and Its Limits

Silver's forward curve is, in normal conditions, governed by broadly the same carry framework described for gold — financing cost less lease rate — but is more susceptible than gold's curve to deviating from a pure carry relationship when industrial-demand-driven physical tightness becomes significant, reflecting the by-product supply inelasticity described in Section 5.

7.1 Normal Contango

- As with gold, in normal market conditions silver trades in contango, with the magnitude of the contango determined primarily by the prevailing USD short-term interest rate less the silver lease rate, following the same formula structure described in detail in the COMEX Gold reference document.

7.2 Backwardation as a More Frequent and More Industrially-Driven Signal Than in Gold

- While backwardation is, as described for gold, a rare and extreme stress signal in that market, silver backwardation, while still an atypical condition, has historically occurred somewhat more readily than gold backwardation, reflecting the combination of: (1) the industrial-demand component that can create genuine near-term physical scarcity in a way gold's almost entirely investment/monetary demand base does not; and (2) the by-product mine supply structure (Section 5.1), which limits the market's ability to rapidly increase mined output in response to a price spike, in contrast to gold, where (though still subject to long lead times) the primary economic driver of most mine investment is the gold price itself.

7.3 Forward Curve Zones

Zone	Contracts	Primary Driver	Secondary Driver	Key Signal
Ultra-front	Spot vs first active	COMEX-LBMA EFP; COMEX registered stock level relative to open interest	Physical retail/industrial demand spikes (as in the 2021 episode, Section 6.1)	Daily EFP; registered stock trend
Front	C1-C4	Short-term USD rates (SOFR); near-term real rate expectations (2-year TIPS)	Silver lease rates; industrial demand data (PMI); ETF flow direction	2-year TIPS; weekly SLV flows; Manufacturing PMI
Belly	C4-C8	Medium-term monetary policy path; 10-year real rate expectations	Solar installation forecasts; mine supply guidance from major by-product producers (Mexico, Peru, Chile)	10-year TIPS; solar industry installation forecasts; major producer guidance
Back	C8+	Long-run real rate expectations; structural industrial demand growth (solar, electronics)	Long-run base metals mining investment cycle (since most silver supply is tied to it); thriving trend in solar	Long-run solar/EV-adjacent demand forecasts; base metals mining capex cycle

7.4 Silver Butterflies

- As with gold, silver butterflies on the COMEX curve are primarily an expression of the curvature of the short-term interest rate path (the same FOMC-meeting-calendar dynamic described for gold butterflies in the COMEX Gold reference document) in normal market conditions, but are more likely than gold butterflies to show a meaningful industrial-demand-driven distortion around a contract month with a specific known physical-demand or supply-disruption catalyst (for example, a major by-product mine disruption in Mexico or Peru), given the industrial-demand sensitivity described

8. The Structural Deficit Narrative

Industry data providers and several major silver market participants have published analyses describing a multi-year period of structural silver market deficits — meaning annual fabrication and investment demand exceeding annual mine and recycled supply — drawing down above-ground inventories over time. As with the analogous copper supply-demand gap thesis described elsewhere in this document series, this narrative should be understood as a long-horizon framework rather than a tactical near-term trading signal.

8.1 The Core Argument

- Sources including the Silver Institute (an industry association) and several metals-focused research providers have published multi-year deficit estimates attributing the gap primarily to solar photovoltaic demand growth (Section 4.1) combined with the by-product supply inelasticity described in Section 5.1, which limits the market's ability to increase mine output in response to either a higher silver price or a higher deficit estimate itself, since most mine output decisions are driven by the economics of the associated base metal rather than by silver.
- **The thesis is subject to the same caveats noted for the analogous copper thesis:** deficit estimates depend on assumptions about solar installation growth rates, the pace of silver thrifting in panel manufacturing, and the responsiveness of by-product mine supply to base-metal (not silver) price trends — all of which carry meaningful forecast uncertainty, and the framework operates on a multi-year horizon rather than providing tactical guidance for near-term spread positioning, which remains dominated by the carry, inventory, and short-term industrial/monetary data described elsewhere in this document.

8.2 Above-Ground Investment Stock as the Buffer

- A structural annual deficit, if it persists, is understood by analysts to be met by drawdowns of above-ground investment-grade silver stock (bars, coins, and ETF holdings) rather than by an inability to physically supply the market in the near term — meaning the practical near-term market effect of a persistent deficit is more likely to manifest as a gradual tightening bias in investment-grade inventory levels and ETF holdings than as an acute physical delivery crisis, absent a separate, more acute triggering event (such as the 2021 episode described in Section 6.1).

9. The Gold-Silver Ratio — The Primary Cross-Market Signal

The gold-silver ratio (GSR) — the gold price divided by the silver price — is the single most important cross-market analytical tool for silver, encoding in one number the relative balance between the monetary-asset framework (Section 3) and the industrial-commodity framework (Section 4) described throughout this document.

9.1 Historical Range and Interpretation

- The GSR has historically ranged from approximately 40:1 to 120:1 in modern markets, with a long-run average commonly cited in the range of 60–70:1, though the appropriate "normal" level is itself a subject of ongoing debate among analysts given structural shifts in both metals' monetary and industrial roles over time (as also noted in the COMEX Gold reference document's discussion of this same ratio from gold's perspective).
- **A high GSR** (silver cheap relative to gold, historically above approximately 90–100:1) has, at various points, coincided with periods of weak industrial demand expectations (recessions, sharp manufacturing slowdowns) where gold's safe-haven characteristics are valued more than silver's industrial-demand exposure, or with periods of acute general risk-off behaviour where gold outperforms as "the purer" safe haven.
- **A low GSR** (silver expensive relative to gold, historically below approximately 50:1) has, at various points, coincided with periods of strong industrial demand and economic optimism, where silver's industrial exposure and historically higher volatility amplify its performance relative to gold during a broadly positive macro and commodity-demand environment.

9.2 The Ratio as a Relative Value Trading Instrument

- Because the GSR oscillates within a historically observed range rather than trending indefinitely in one direction, it is sometimes used by precious metals traders as a tactical mean-reversion signal — taking a position that is long the historically cheap metal and short the historically expensive one on a ratio basis, rather than taking an outright directional view on either metal's flat price. This approach treats the GSR level itself, rather than either metal's absolute price level, as the primary trading signal.

9.3 The Ratio as a Macro Regime Indicator

- Beyond its tactical trading use, the GSR's trend direction over a period of weeks to months is also used by analysts as a broader real-time signal of whether market participants are, in aggregate, weighting the macro/monetary narrative (gold outperforming, ratio rising) or the industrial-demand/risk-on narrative (silver outperforming, ratio falling) more heavily at a given point in time — providing a useful cross-check against the specific data points described in Sections 3 and 4 of this document.

10. Other Cross-Market Relationships

10.1 Silver and Copper

- Given the by-product mine supply relationship described in Section 5.1 (a meaningful share of silver supply is tied to copper mining economics) and silver's own industrial demand profile (Section 4), silver exhibits a documented, if imperfect, positive correlation with copper — both metals respond to global manufacturing and industrial activity data, and both are affected by overlapping base-metals-complex sentiment and Chinese demand-cycle dynamics described in the COMEX Copper reference document.

10.2 Silver and the Broader Base Metals Complex

- Beyond copper specifically, silver's by-product relationship to lead-zinc mining (Section 5.1) means that lead and zinc market conditions — though not separately covered in detail in this document series — are a relevant background supply variable for silver in a way they are not for gold.

10.3 Silver and Global Manufacturing PMI

- Global and major-economy (US, China, Eurozone) Manufacturing PMI readings are a more directly relevant demand-side indicator for silver than for gold, given silver's substantial industrial fabrication demand component (Section 4) — readings significantly above or below the 50 expansion/contraction threshold are monitored as a general industrial-demand health signal relevant to silver in a way they are generally not directly relevant to gold pricing.

10.4 Speculative Positioning — COT Data

- CFTC COT for COMEX Silver:** weekly managed money net positioning data is closely tracked, following the same general crowding/reversal-risk interpretive framework described throughout this document series. Silver COT positioning has historically been associated with periods of pronounced retail and momentum-driven speculative interest (including the episode referenced in Section 6.1), making positioning extremes arguably even more closely watched in silver than in some other metals markets covered in this series.

11. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Release Timing	Primary SI Curve Impact
US CPI / Core PCE	BLS / BEA	Monthly	Per standard release calendar	All contracts — shared with gold via the real-rate/monetary framework (Section 3)
FOMC rate decision + SEP	Federal Reserve	8× per year (SEP 4× per year)	2:00 PM ET	All contracts — shared with gold; carry structure repricing
10-year TIPS yield (continuous)	Bloomberg / Reuters / Fed	Continuous	Market hours	All contracts — monetary-framework input shared with gold
US Dollar Index (DXY)	ICE / Bloomberg	Continuous	Market hours	All contracts — shared negative-correlation mechanism with gold
Global / US / China / Eurozone Manufacturing PMI	S&P; Global / ISM / Caixin / others	Monthly	1st-5th of month (varies by region)	C2-C8 — silver-specific industrial demand signal with no direct gold equivalent
SLV (iShares Silver Trust) daily holdings	BlackRock	Daily	After close / next morning	C1-C4 — investment demand signal, analogous to GLD for gold
COMEX registered/eligible silver stocks	CME Group	Daily	After close	Front contract — delivery squeeze risk; physical tightness signal
LBMA Silver Price	IBA via LBMA	Daily (single fix)	~12:00 London	Spot reference — note single daily fix, unlike gold's AM/PM structure
Silver Institute World Silver Survey	Silver Institute / Metals Focus	Annual + periodic updates	Typically published Q1/Q2	C8+ — comprehensive annual supply/demand balance; the primary structural-deficit-narrative source
Global solar PV installation data and forecasts	BNEF, IEA, SolarPower Europe, and others	Periodic / annual	Variable	C6+ — the key structural industrial demand growth input
Major by-product producer quarterly reports (Mexican, Peruvian, Chilean miners)	Fresnillo, Peñoles, Hochschild, and others; also major copper/lead-zinc miners with silver by-product	Quarterly	Varies by company	C3-C12 — mine-level guidance and disruption disclosure
CFTC Commitments of Traders (Silver)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — historically prone to sharper extremes than gold COT
LBMA London vault silver holdings	London Bullion Market Association	Monthly	Variable	C4-C12 — broader above-ground investment-grade stock signal beyond COMEX-specific data

COMEX Silver is best understood as a metal that must be analysed through two simultaneously operating frameworks rather than one: the real-interest-rate and USD-driven monetary-asset framework that dominates gold, and an industrial-demand and by-product-supply framework that has more in common with copper and the base metals complex. Neither framework alone explains silver's price behaviour reliably; the skill in trading silver spreads lies in correctly weighing which framework is dominant at a given moment, using the gold-silver ratio as the primary real-time signal of that balance.

For spread traders, the most actionable signals are: the carry structure (USD rates less silver lease rate) that governs the curve in normal conditions, much as in gold; the by-product supply inelasticity (Section 5.1) that can produce sharper and more persistent backwardation than gold when industrial demand or a mine disruption creates genuine physical tightness; the solar photovoltaic installation and thrifting data that drives the structural deficit narrative's credibility at any given time; and the gold-silver ratio itself, both as a tactical relative-value instrument and as a continuous read on which of silver's two identities the market is currently pricing more heavily.